

SOCIAL MEDIA



MENTAL HEALTH

TASKS + USERS + DATA



General Task

Interactive dashboard exploring social media usage and emotional well-being

Users

- Social Media Users
- Social Media Analysts

Data Source

Kaggle: Social Media Usage and Emotional Well-Being



TOOLS AND TECHNOLOGIES



Tableau



Bootstrap



HTML



JavaScript



CSS

INDIVIDUAL TASKS



Task 1 - Demographics

Insights in how attributes (age, gender, etc.) affect social media use.

Task 2 - Platforms

Determine what emotions dominate which platform.

Task 3 - Time Spent

Discover how time on social media affects user emotions.

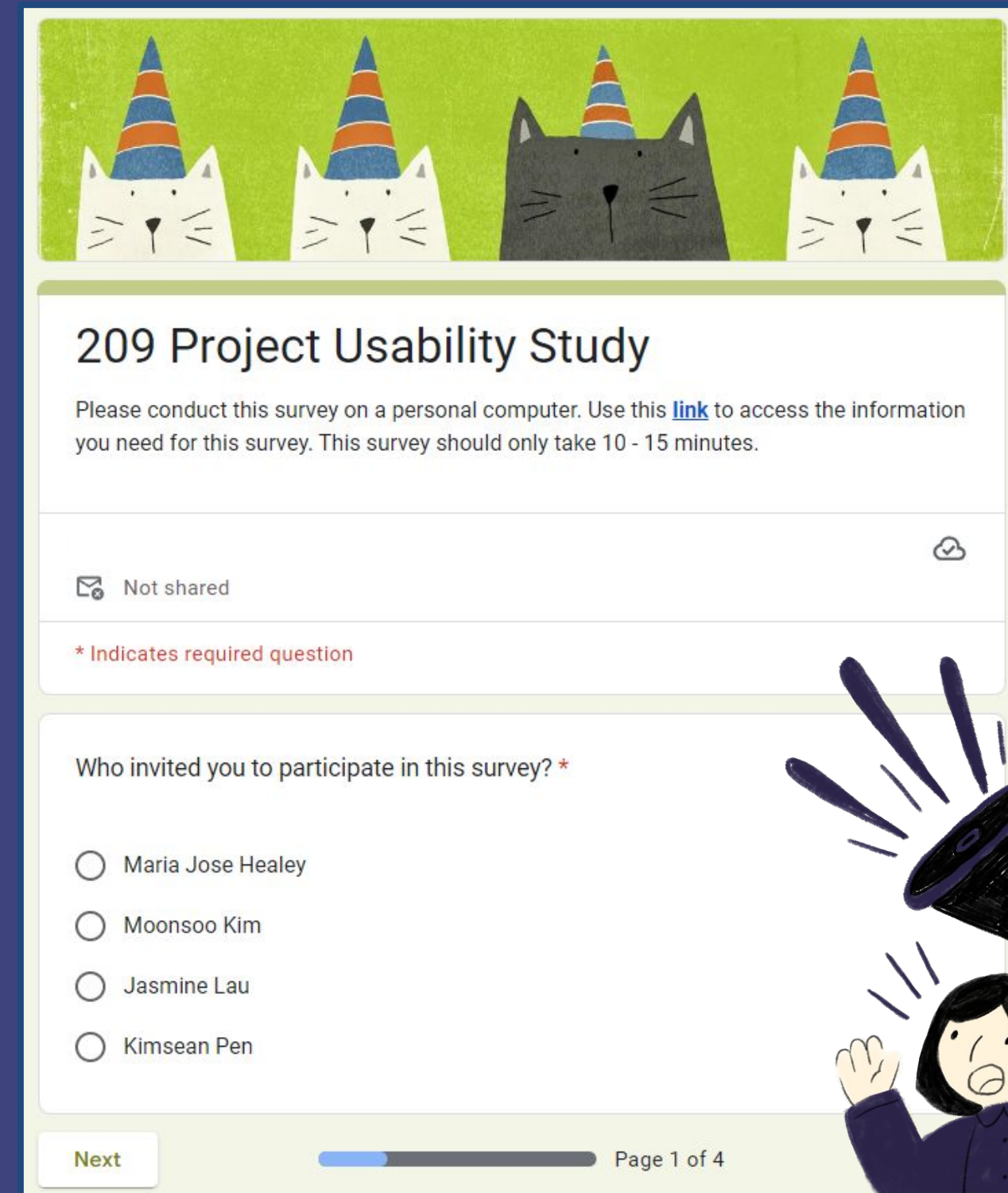
Task 4 - Interactions

Explore how emotional intensity correlates with daily social media use.

USABILITY STUDY DEBRIEF

Survey:

- Questions revolved around each analysis section
- Overall:
 - Need for distinct titles
 - More consistent color encoding
 - Utilize drop down menus
 - Horizontal layout > Vertical layout
 - Reduce mental load
 - Provide more context



The screenshot shows a survey interface. At the top is a header with four cartoon cats wearing party hats. Below the header is the title '209 Project Usability Study' and a paragraph of instructions. A status bar indicates 'Not shared'. A legend states '* Indicates required question'. The main question is 'Who invited you to participate in this survey? *' with four radio button options. At the bottom is a 'Next' button, a progress bar, and 'Page 1 of 4'.

209 Project Usability Study

Please conduct this survey on a personal computer. Use this [link](#) to access the information you need for this survey. This survey should only take 10 - 15 minutes.

Not shared

* Indicates required question

Who invited you to participate in this survey? *

☐ Maria Jose Healey

☐ Moonsoo Kim

☐ Jasmine Lau

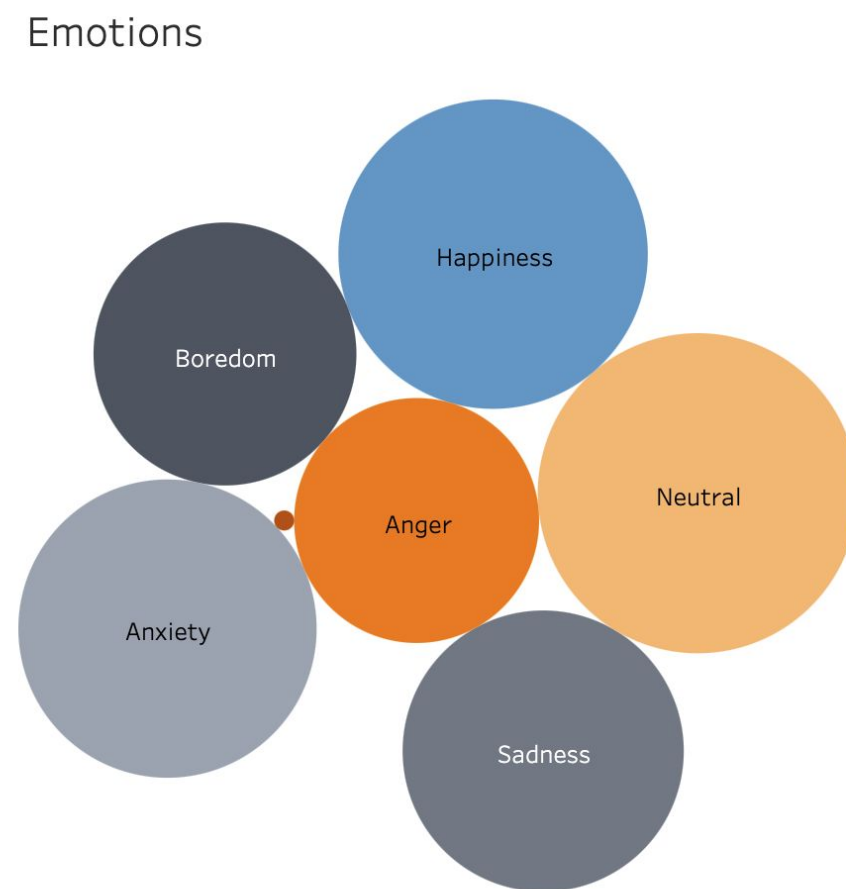
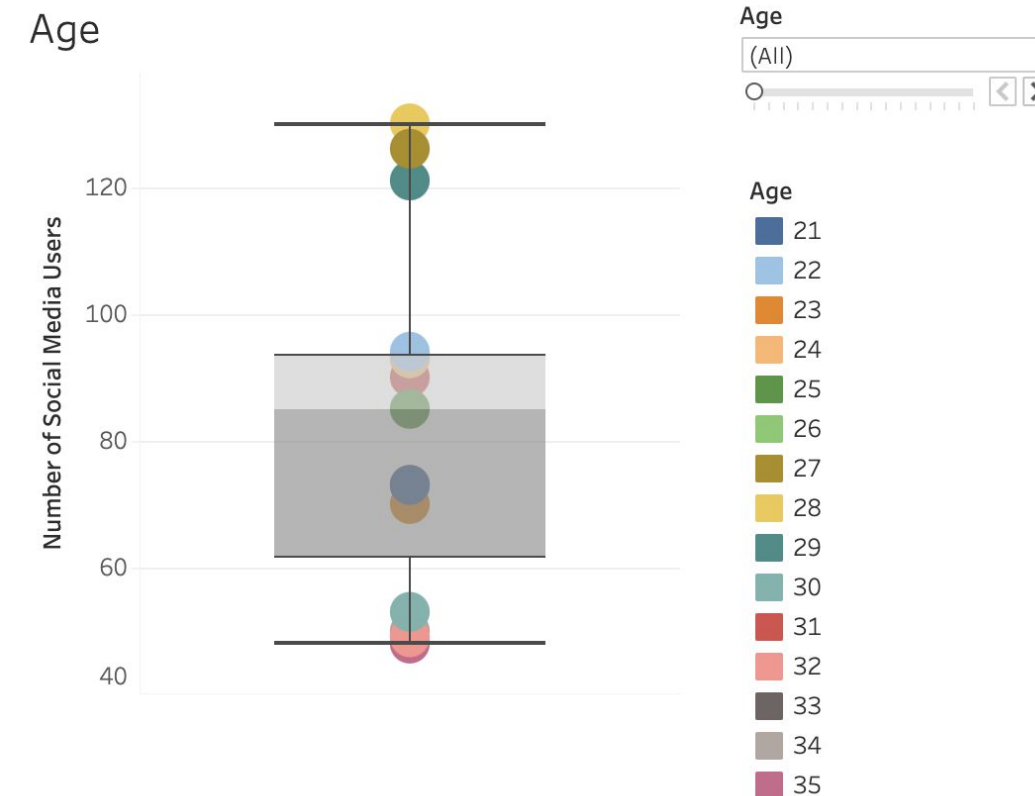
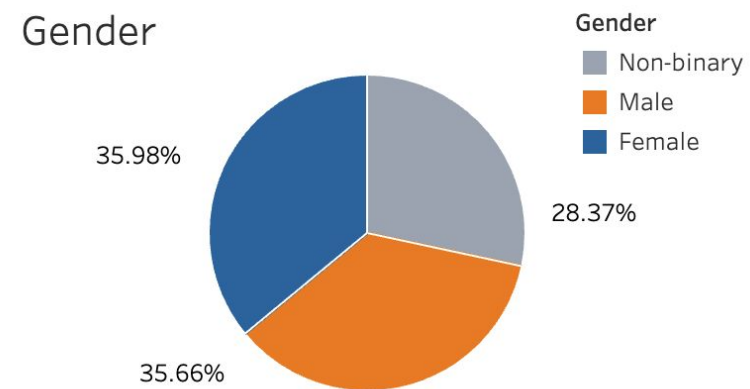
☐ Kimsean Pen

Next

Page 1 of 4



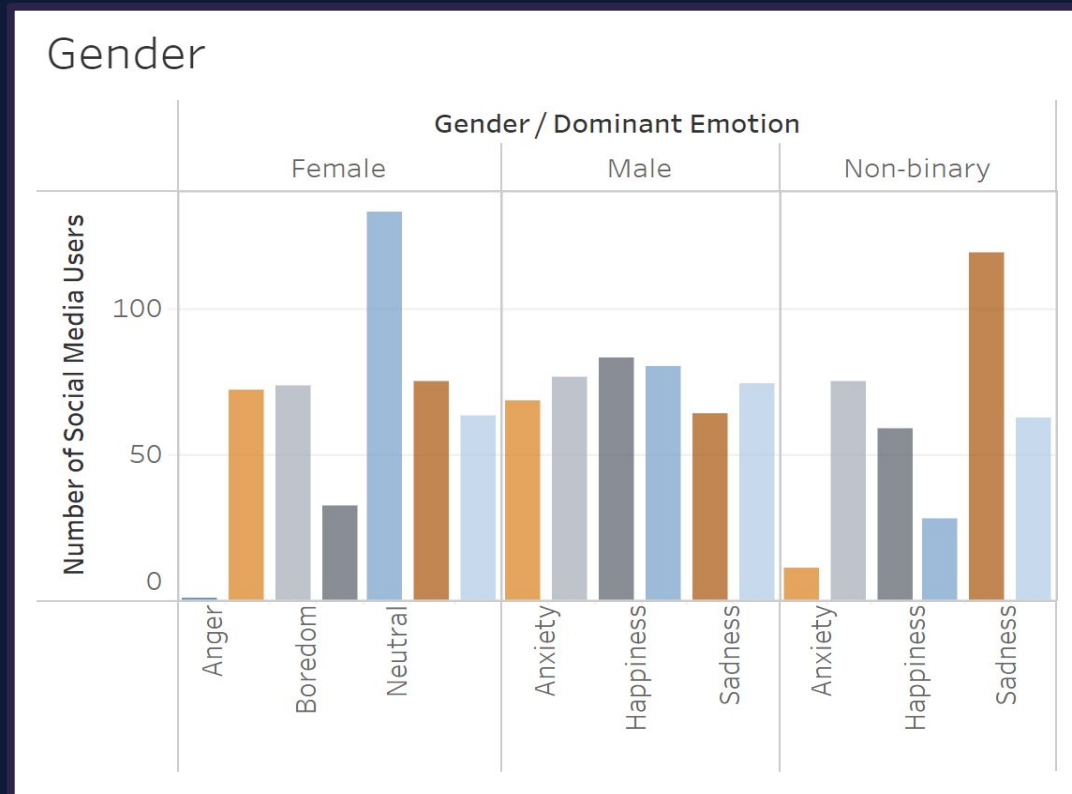
TASK 1 - DEMOGRAPHICS



- pie chart
 - percentage and quantity of each gender in the dataset
- box-and-whisker plot
 - spread of distribution of users' age
- packed bubbles
 - clusters of emotions

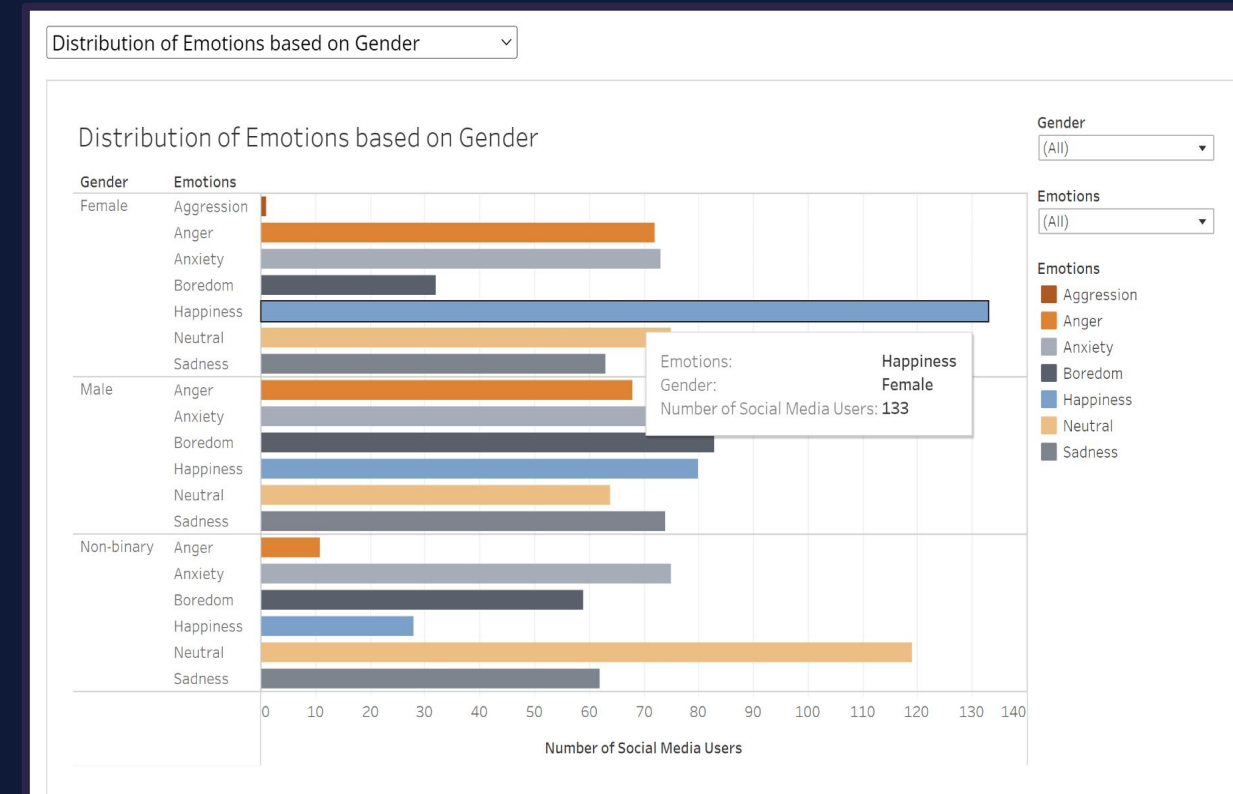
TASK 1 - DISTRIBUTION BY GENDER

before - vertical bars



- visibility affected by multiple charts on same dashboard
- vertical orientation of bars
- no interactivity

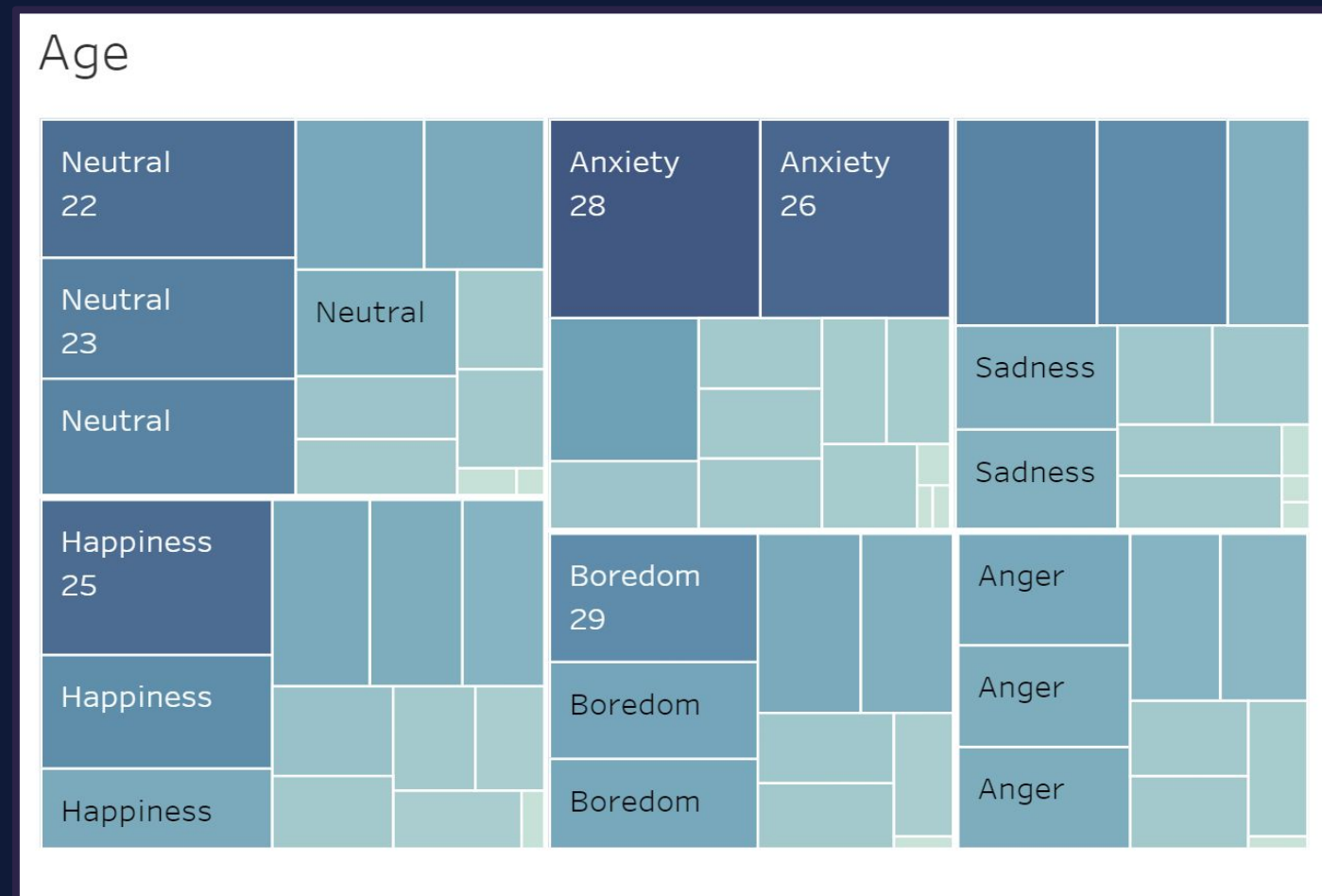
after - horizontal bars



- visibility improved by single chart on each dashboard
- horizontal orientation of bars
- interactivity: slider, drop down

TASK 1 - DISTRIBUTION BY AGE

before - treemap



- not easily understood
- not intuitive / unfamiliar
- color inaccessible

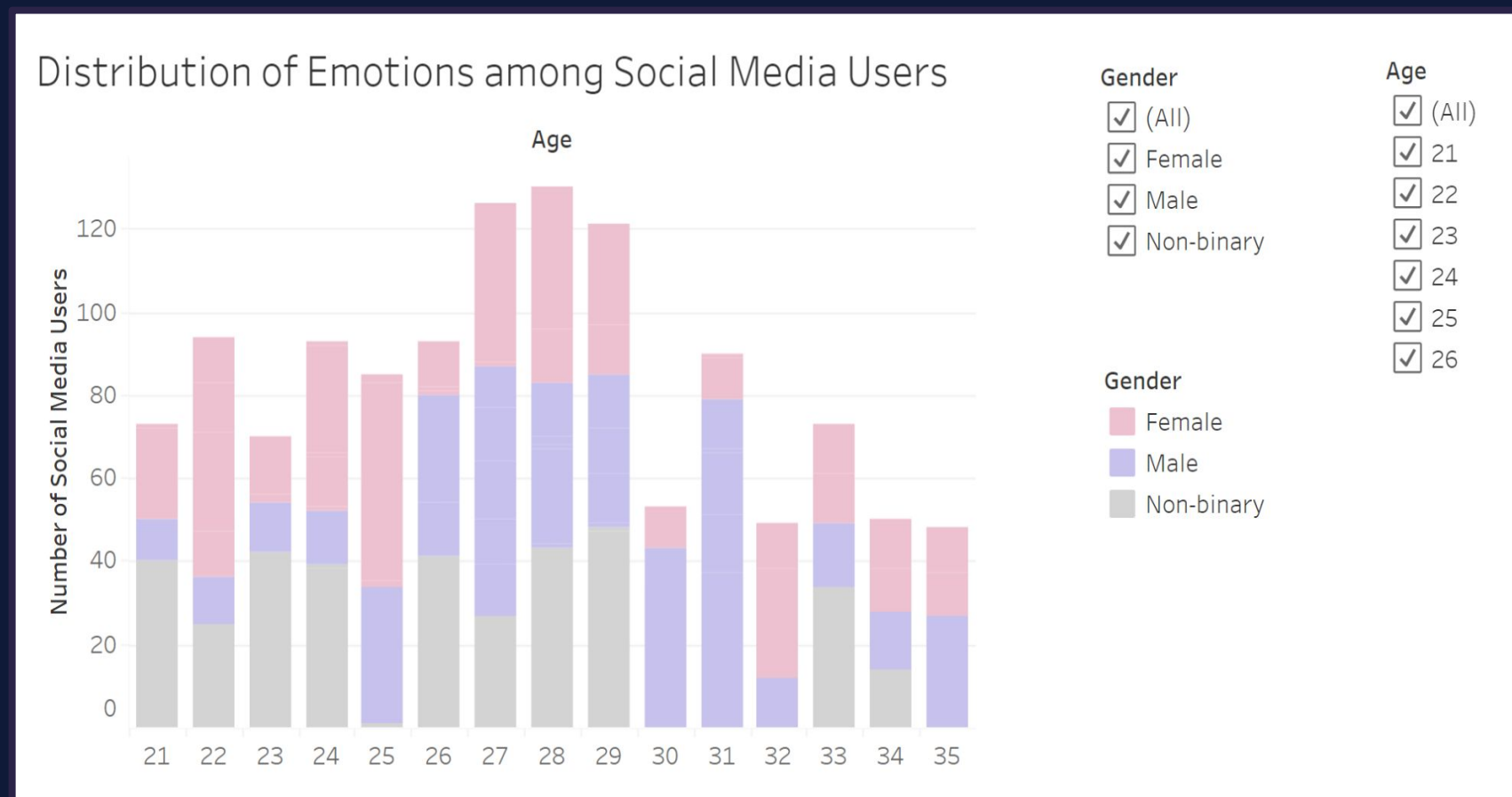
after - scatter plot



- clear relationships
- interactive
- color accessible

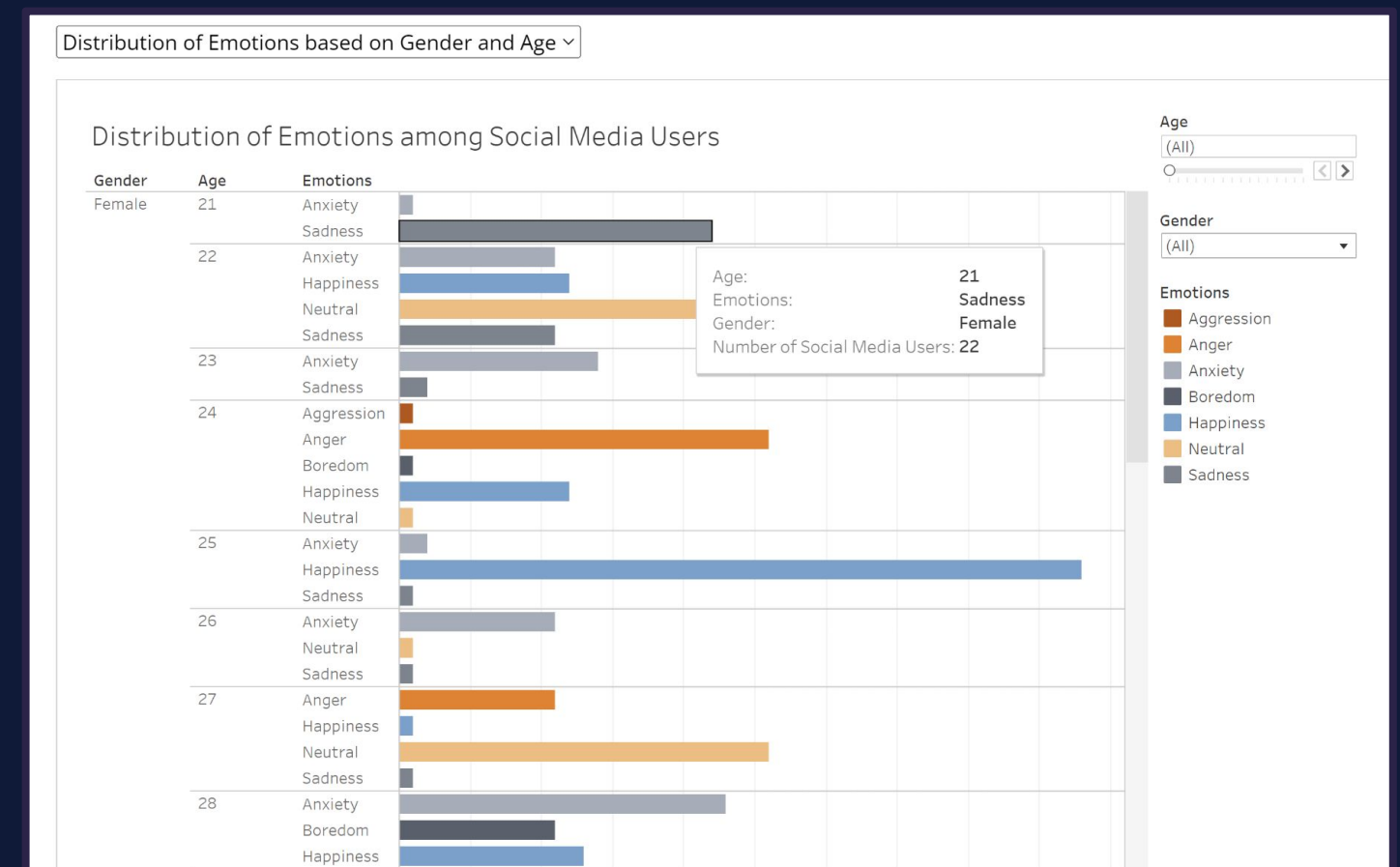
TASK 1 - DISTRIBUTION BY GENDER & AGE

before - stacked bar chart



- not easily operable
- vertical orientation of stacked bars
- filters

after - horizontal bar chart



- straightforward
- horizontal orientation of bars
- interactivity: slider, drop down menu

TASK 2 - PLATFORM

heatmap

before

Dominant E..	Platform						
	Facebook	Instagram	LinkedIn	Snapchat	Telegram	Twitter	Whatsapp
Aggression		1					
Anger		12	1		11	92	35
Anxiety	66	35	27	32	13	27	24
Boredom	51	1	83		11	28	
Happiness		205		13		13	10
Neutral	92	25	31	28	37	25	20
Sadness	39	24	12	39	26	59	

- words cut off
- cells melded together
- diverging colors do not make sense

after

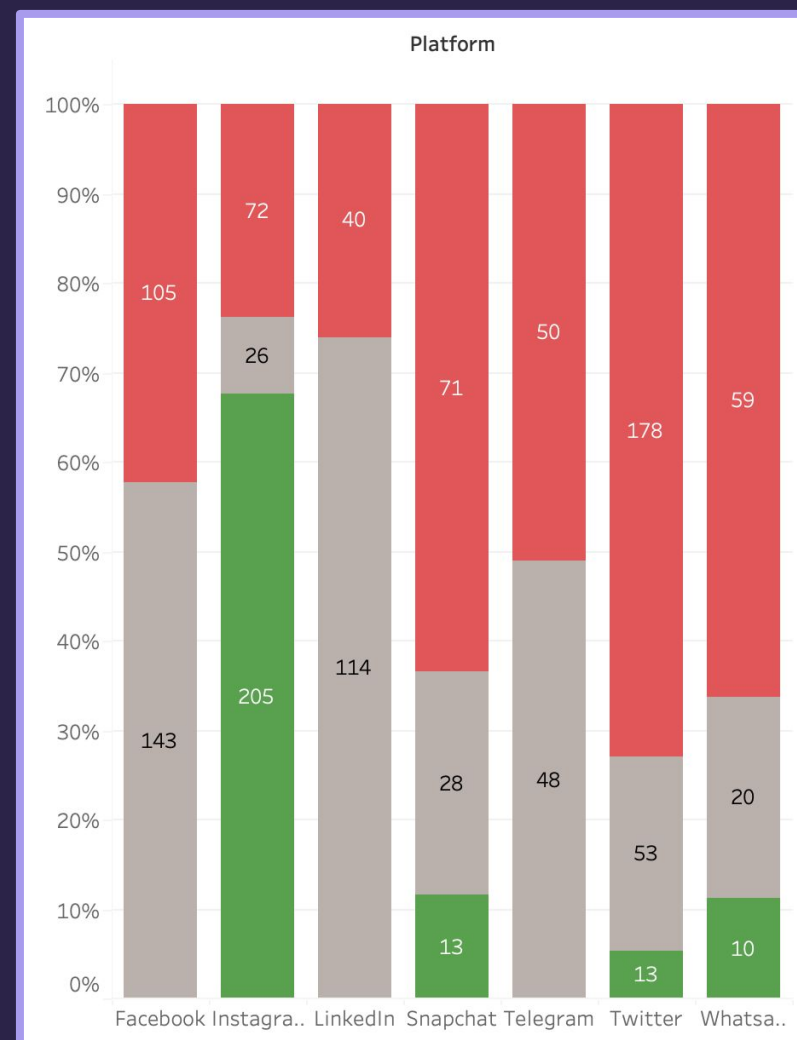
Dominant Emotion	Platform						
	Facebook	Instagram	LinkedIn	Snapchat	Telegram	Twitter	Whatsapp
Aggression		1					
Anger		12	1		11	92	35
Anxiety	66	35	27	32	13	27	24
Boredom	51	1	83		11	28	
Happiness		205		13		13	10
Neutral	92	25	31	28	37	25	20
Sadness	39	24	12	39	26	59	

- labels and annotations are clear
- cells clearly delineated
- gradient coloring is more cohesive

TASK 2 - PLATFORM

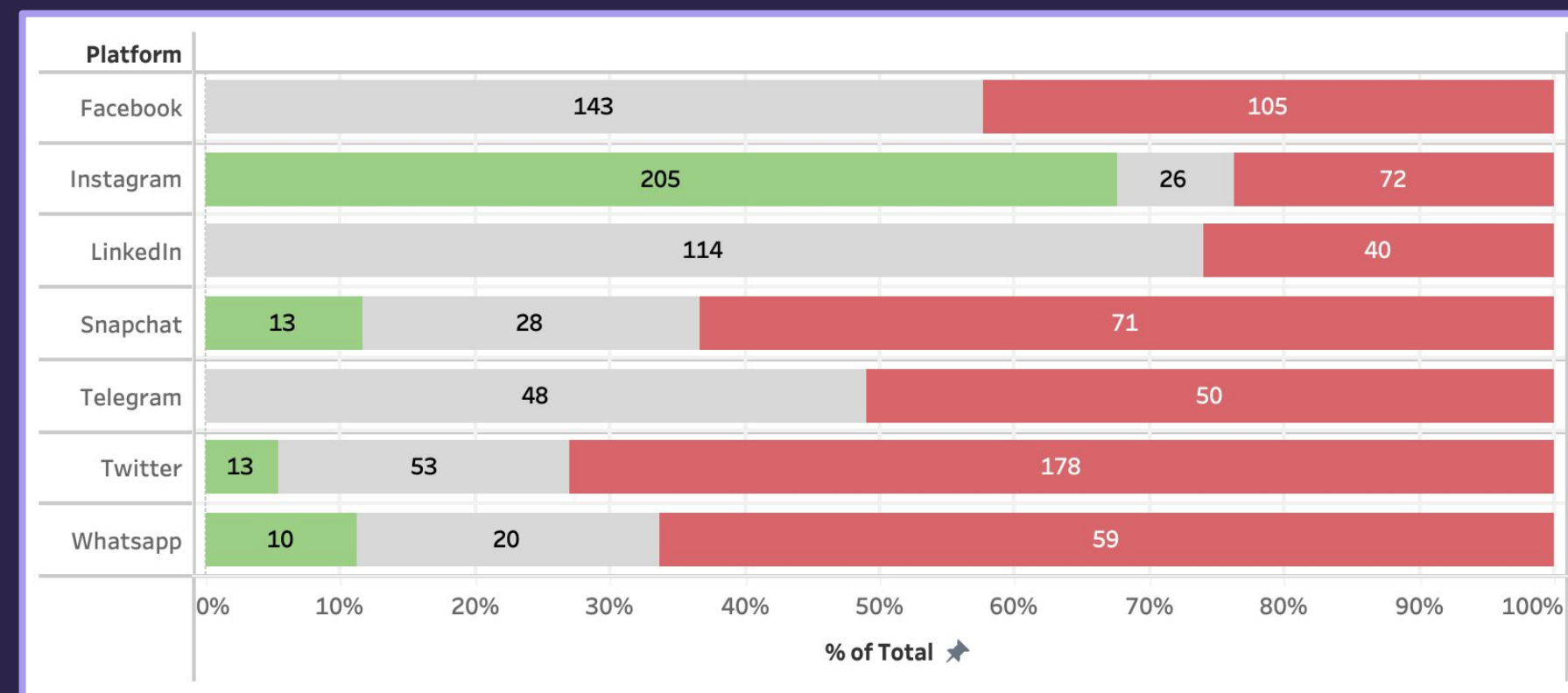
proportional stacked bar chart

before



- vertical orientation
- no axis titles

after

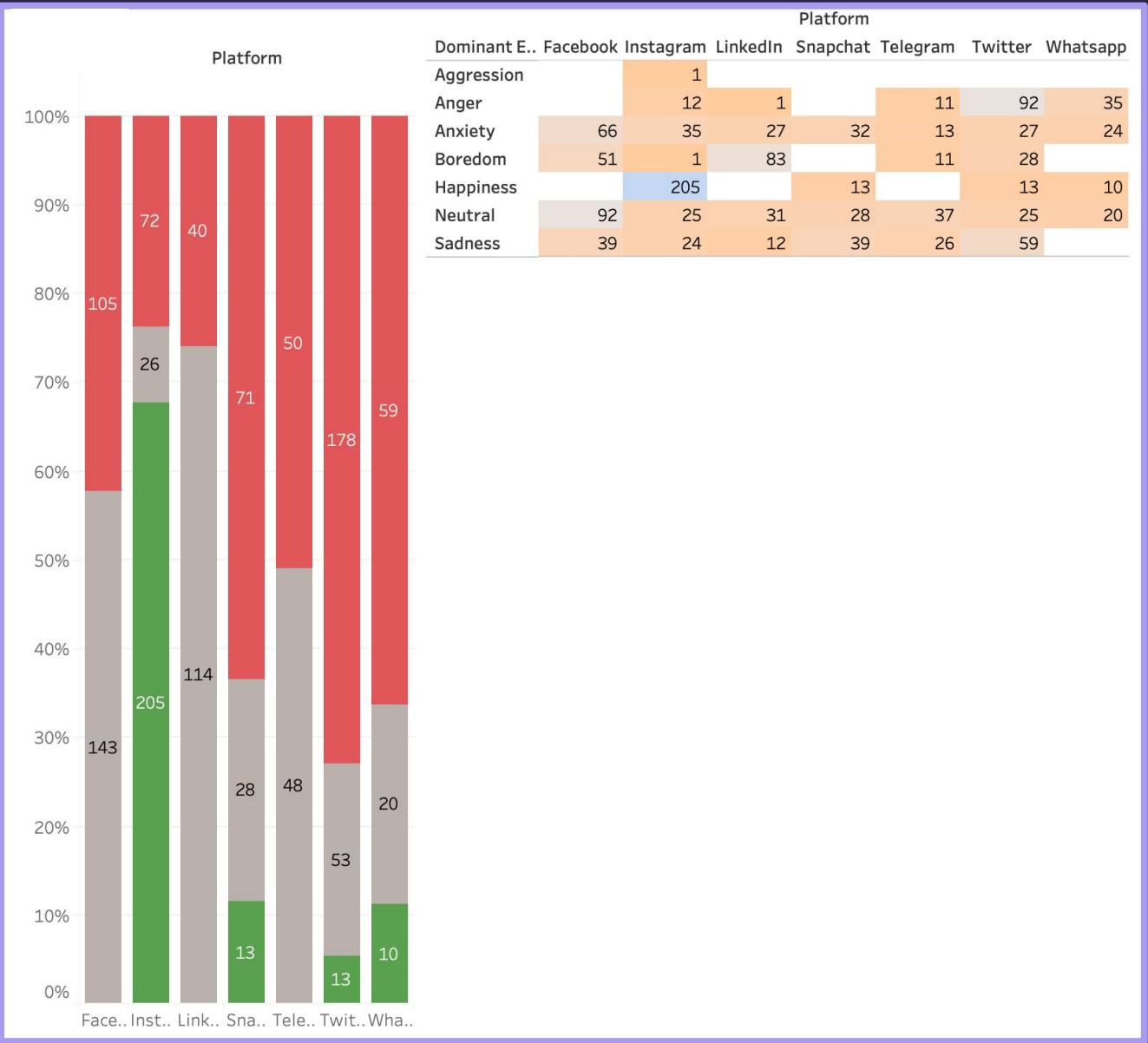


- horizontal orientation
- labels are clear

TASK 2 - PLATFORM

platform dashboard

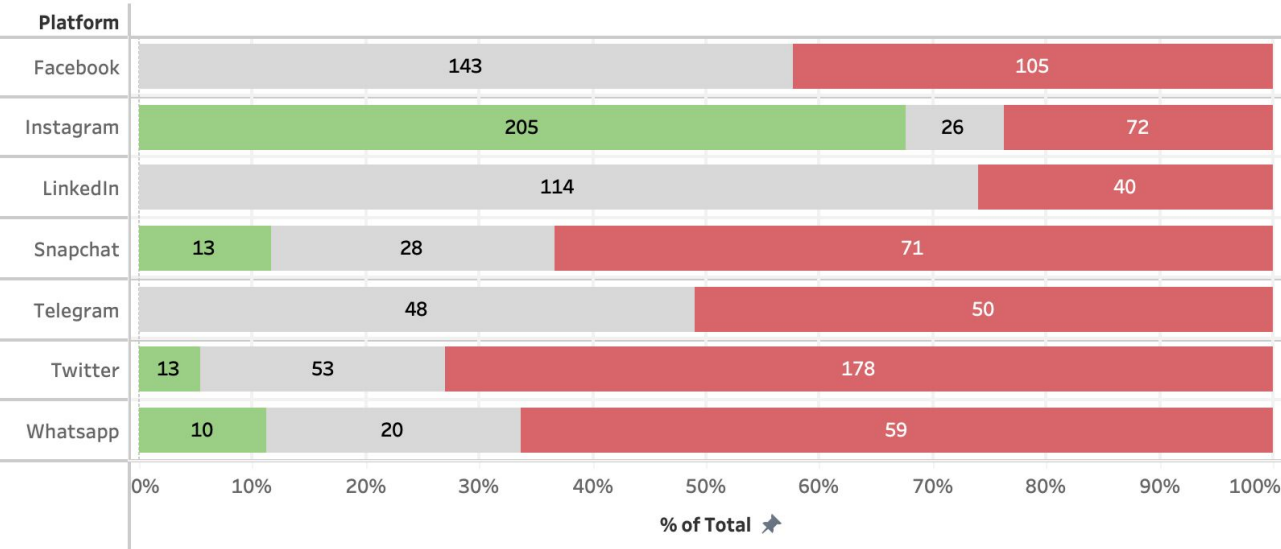
before



- use of space
- amount of whitespace
- clarity of information

after

Proportion of Categorized Emotion by Platform



Number of Users' Emotion by Platform

Dominant Emotion	Facebook	Instagram	LinkedIn	Platform Snapchat	Telegram	Twitter	Whatsapp
Aggression		1					
Anger		12	1		11	92	35
Anxiety	66	35	27	32	13	27	
Boredom	51	1	83		11	28	
Happiness		205		13		13	10
Neutral	92	25	31	28	37	25	20
Sadness	39	24	12	39	26	59	

TASK 3 - TIME SPENT

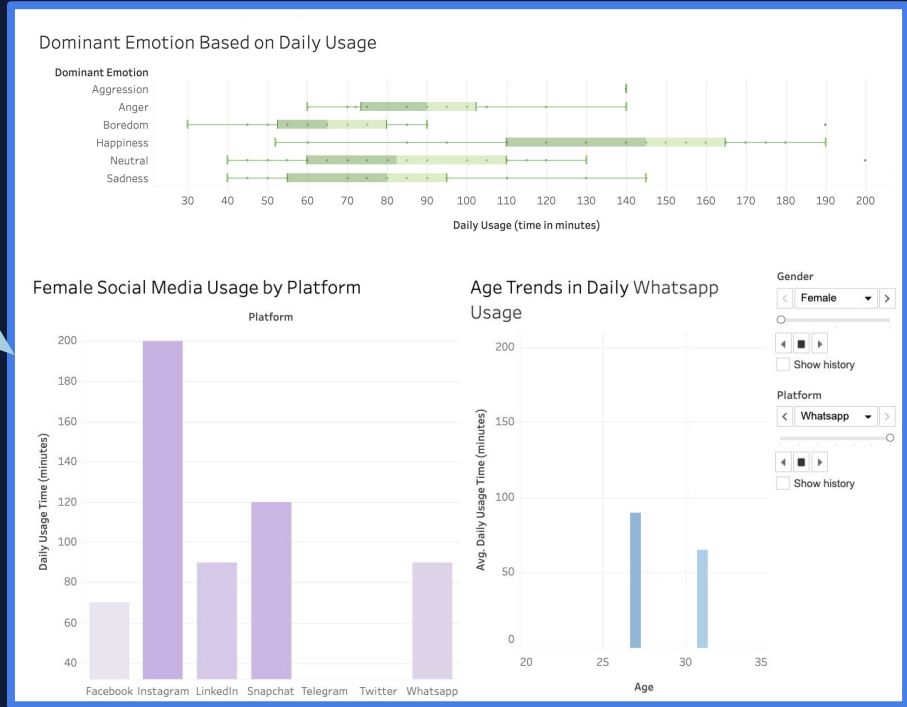
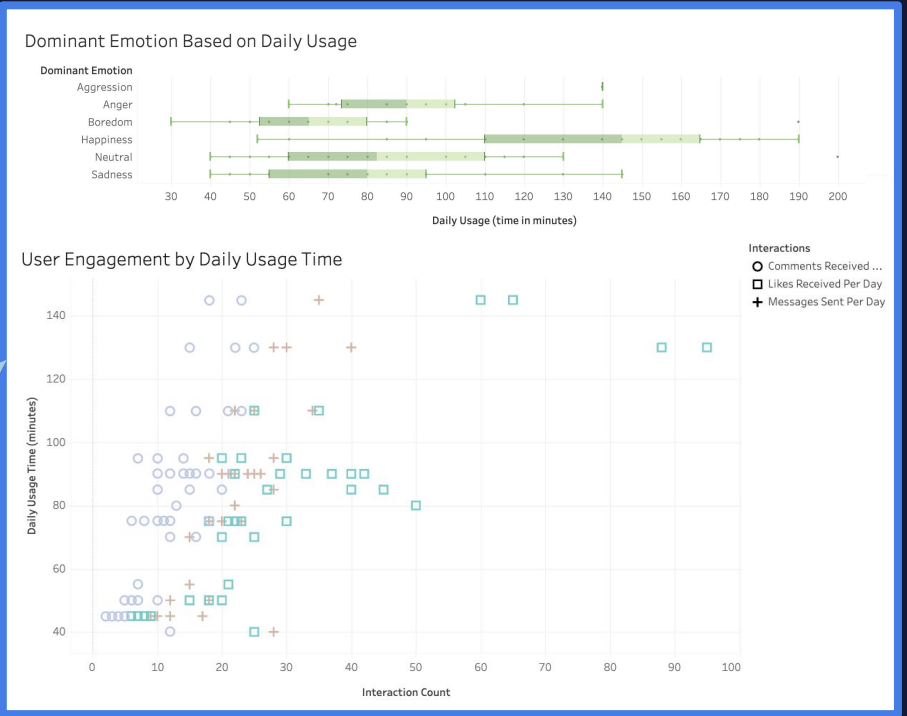
before



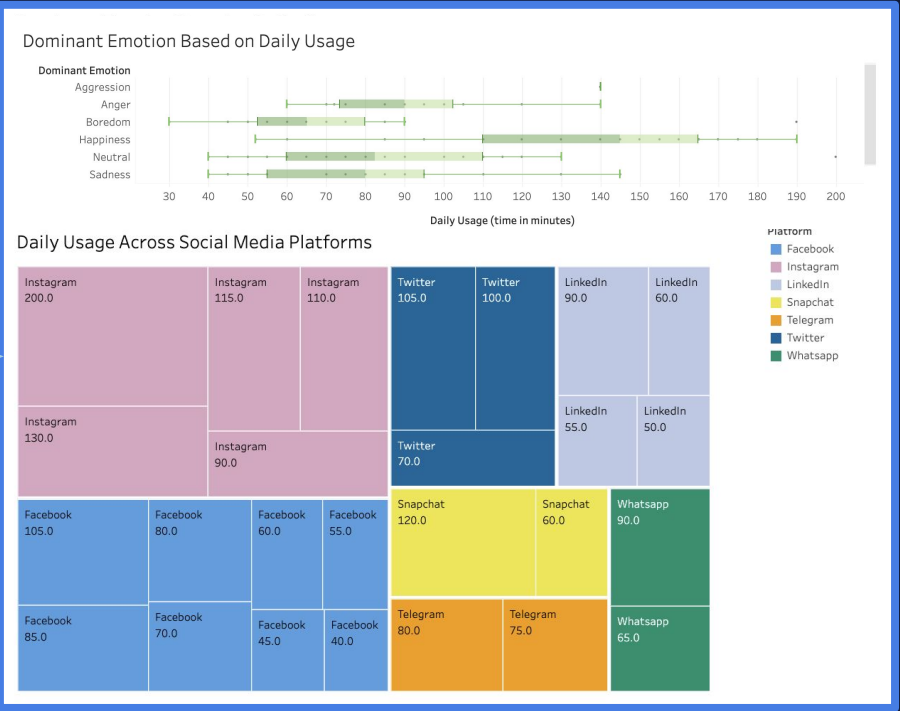
too overwhelming

reducing amount of information in a graph

too much noise



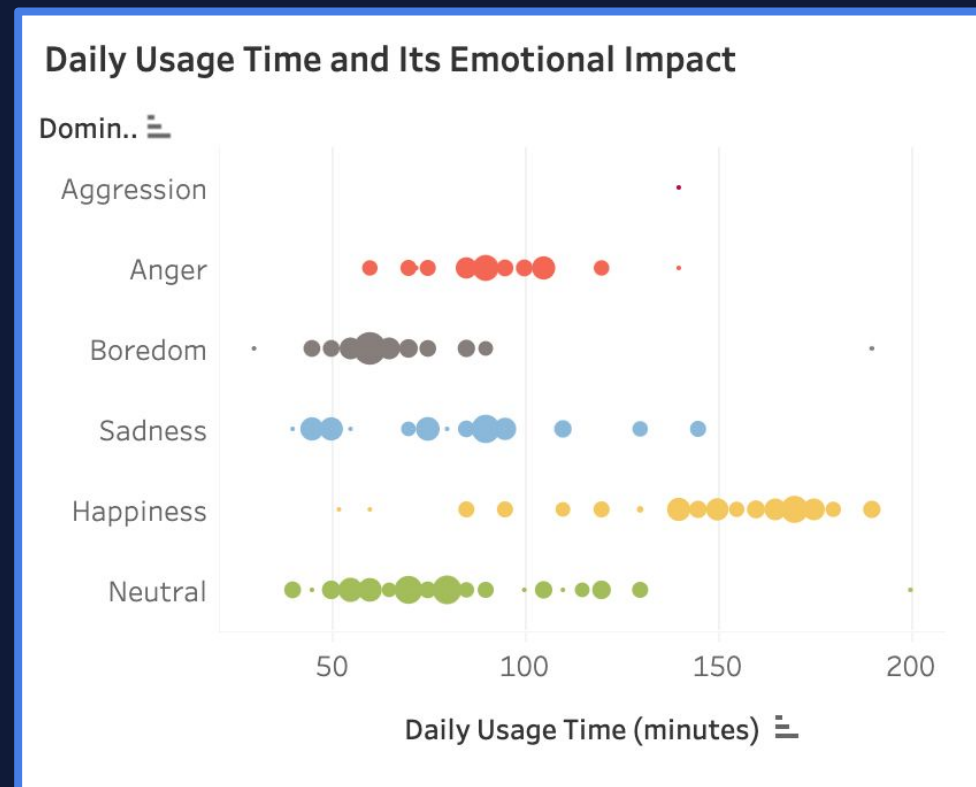
after



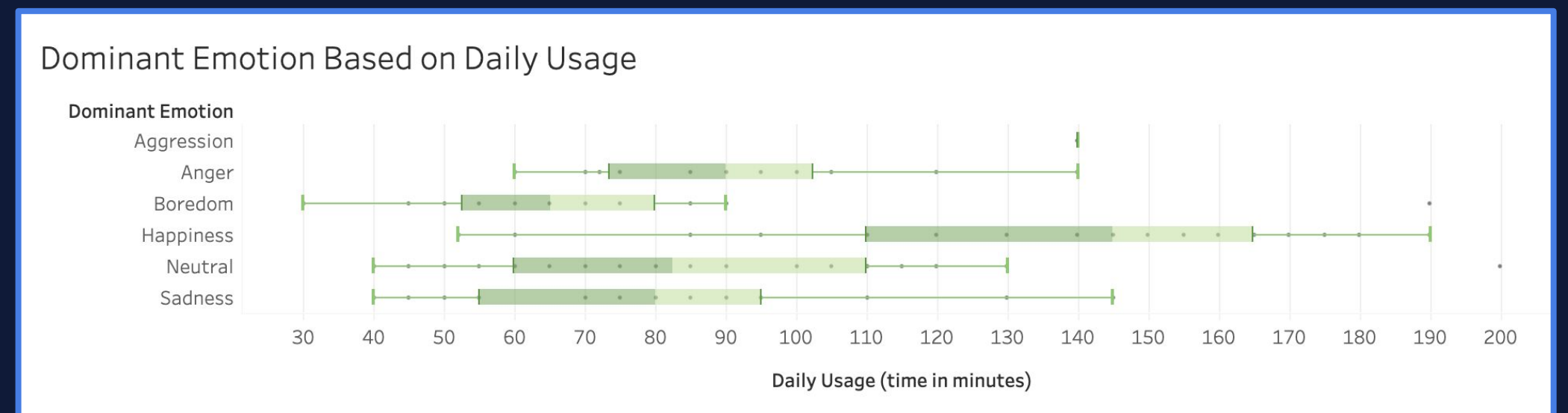
TASK 3 - TIME SPENT

dominant emotion based on daily usage

before



after



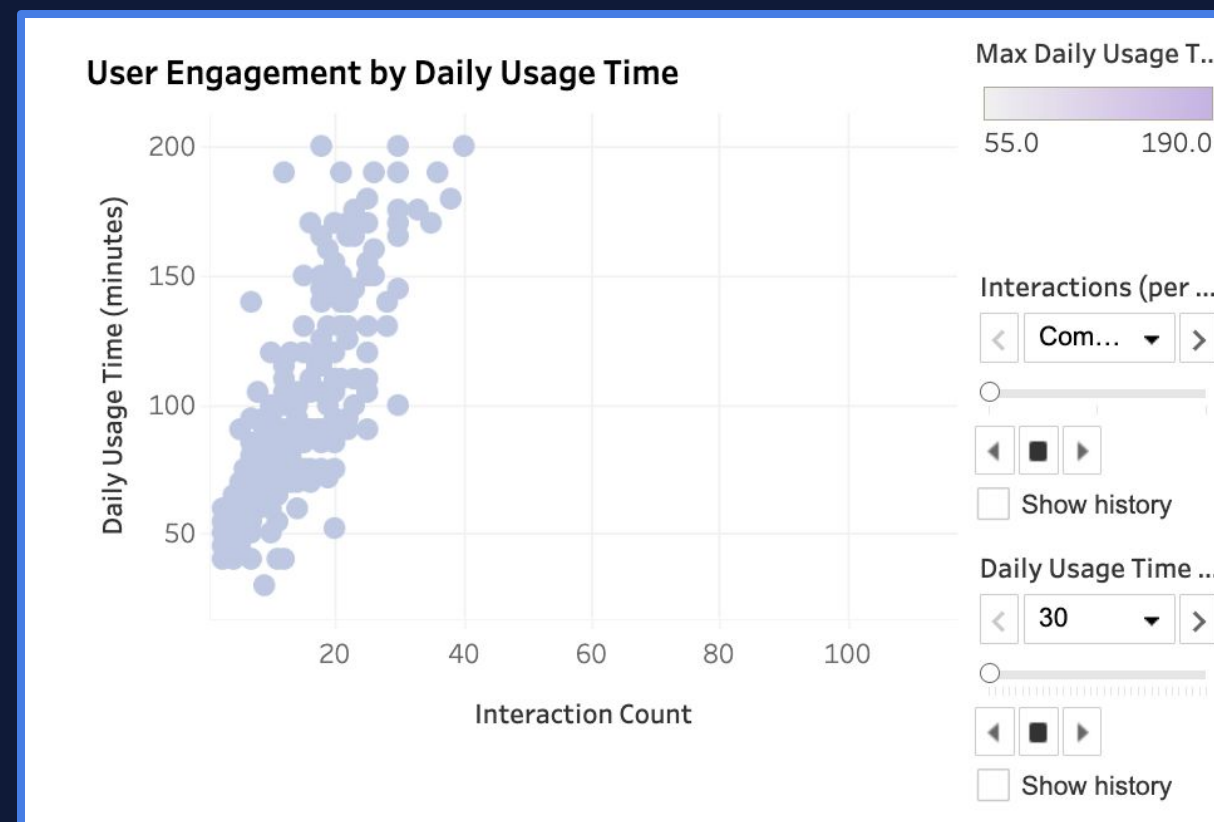
- confusion around dot sizing
- increased noise due to color and size encoding

- less noise
- clear indication which emotion is dominant as usage time ↑
- simple and easy to read

TASK 3 - TIME SPENT

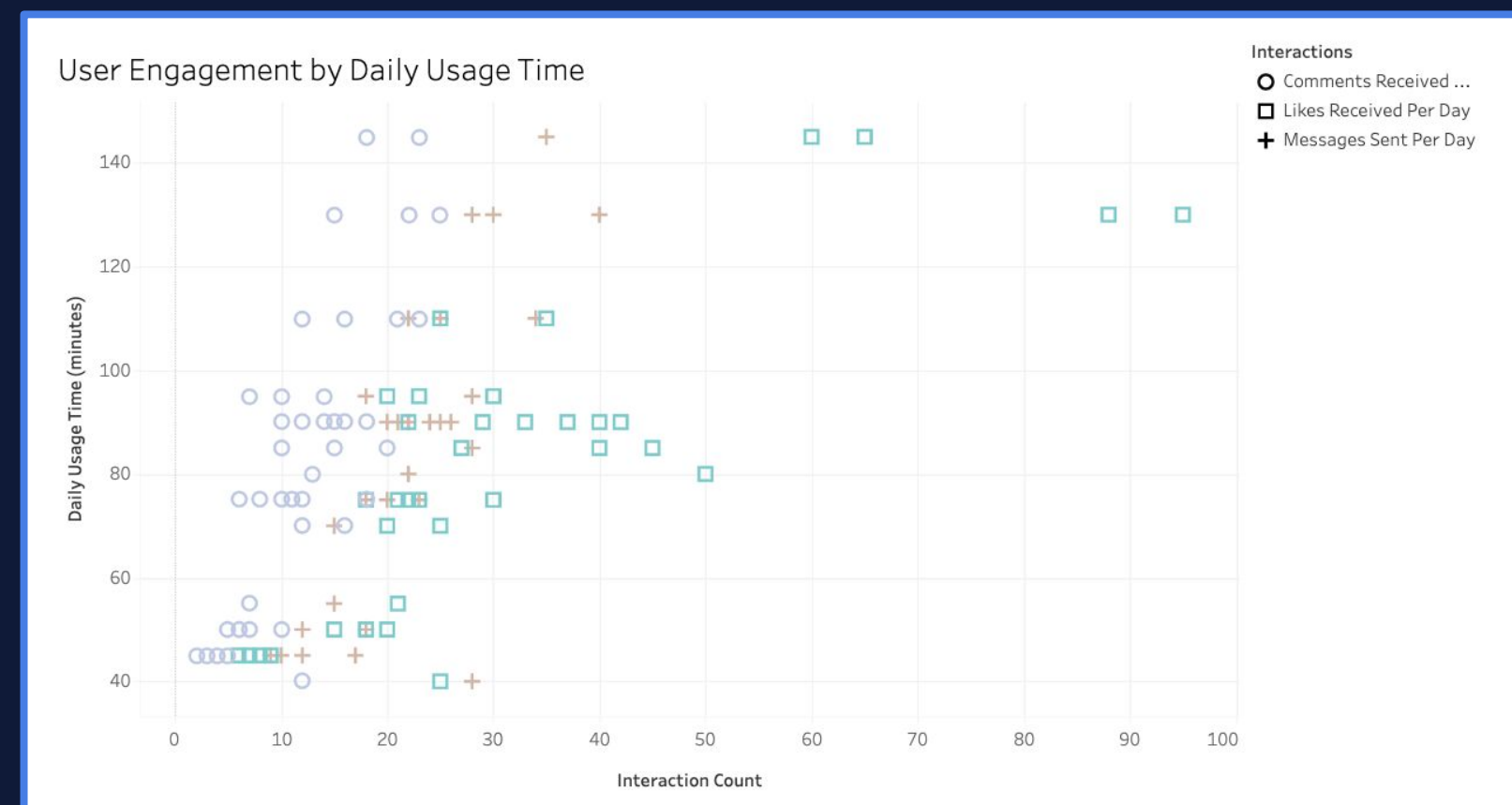
user engagement by daily usage

before



- different graphs for each interaction
 - too many types of interactions
- count scale too large

after

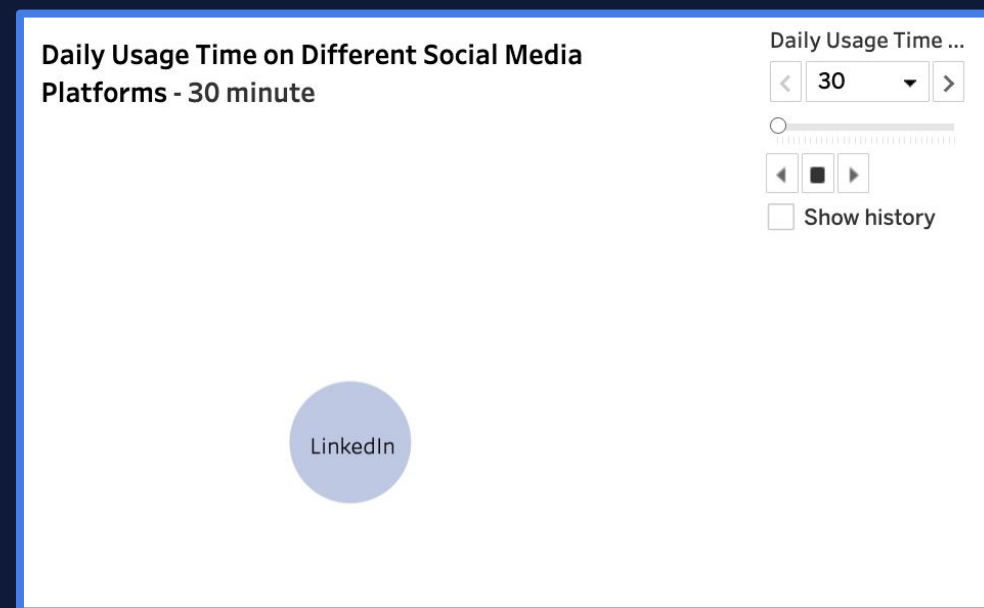


- all condensed into one graph
 - easier comparison between interaction types

TASK 3 - TIME SPENT

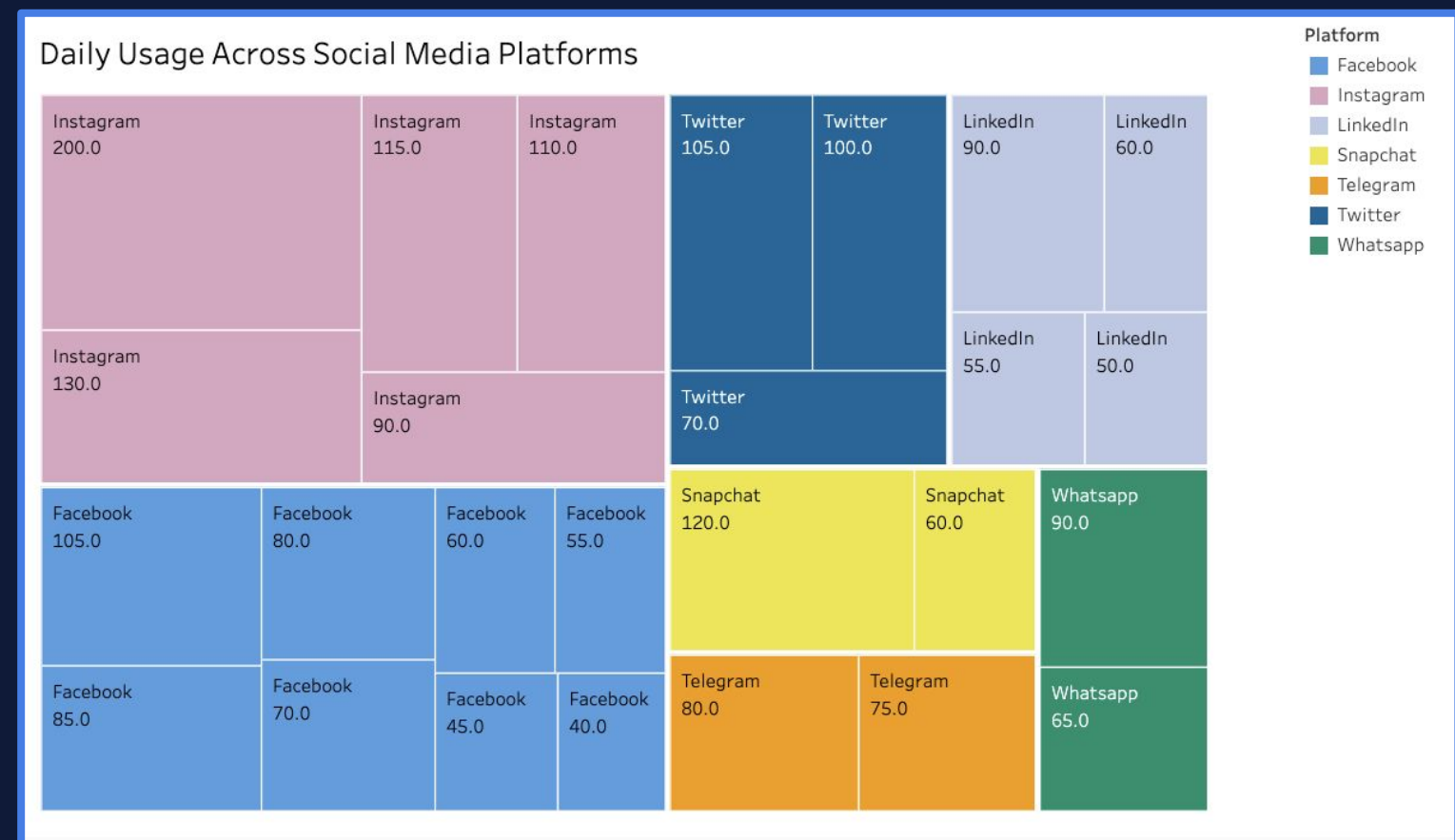
daily usage across social media platforms

before



- 77% answered survey question correctly
 - interactivity was not clear due to overwhelming dashboard

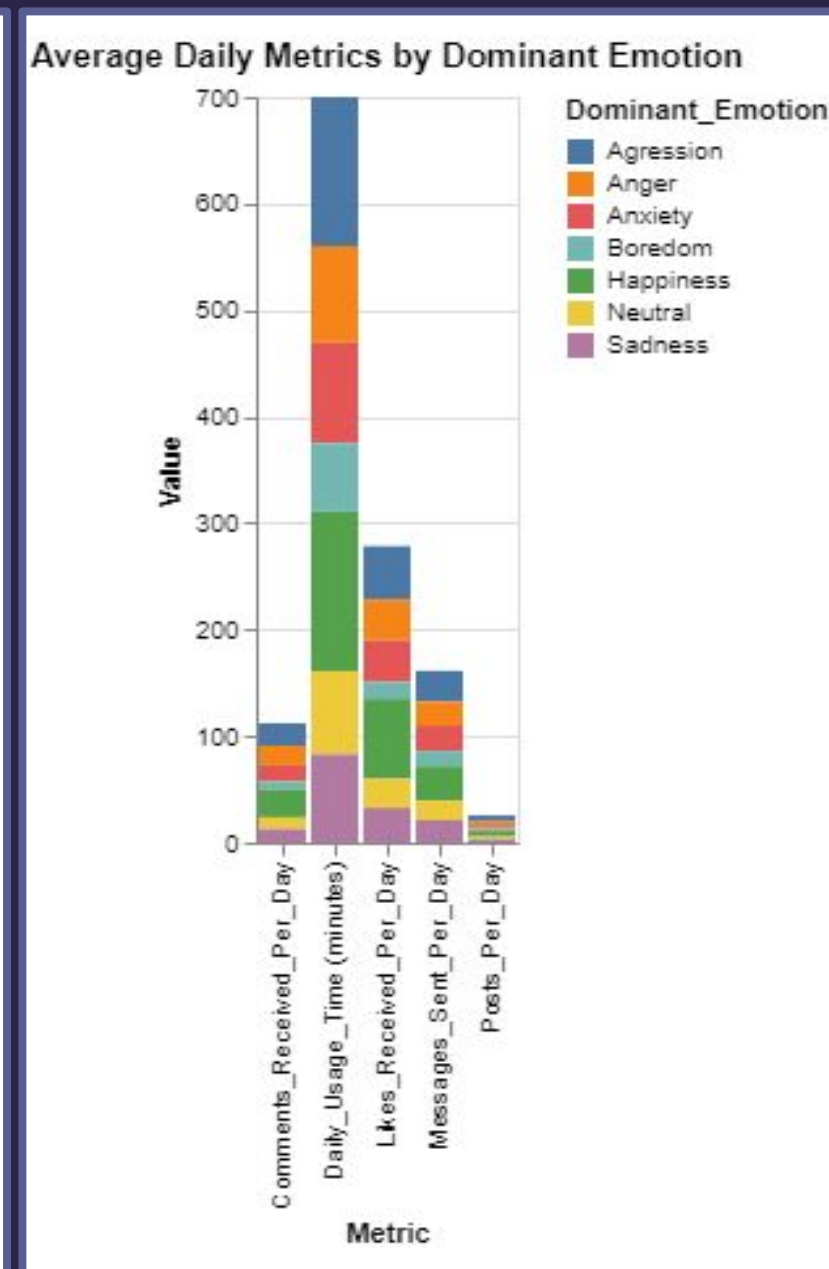
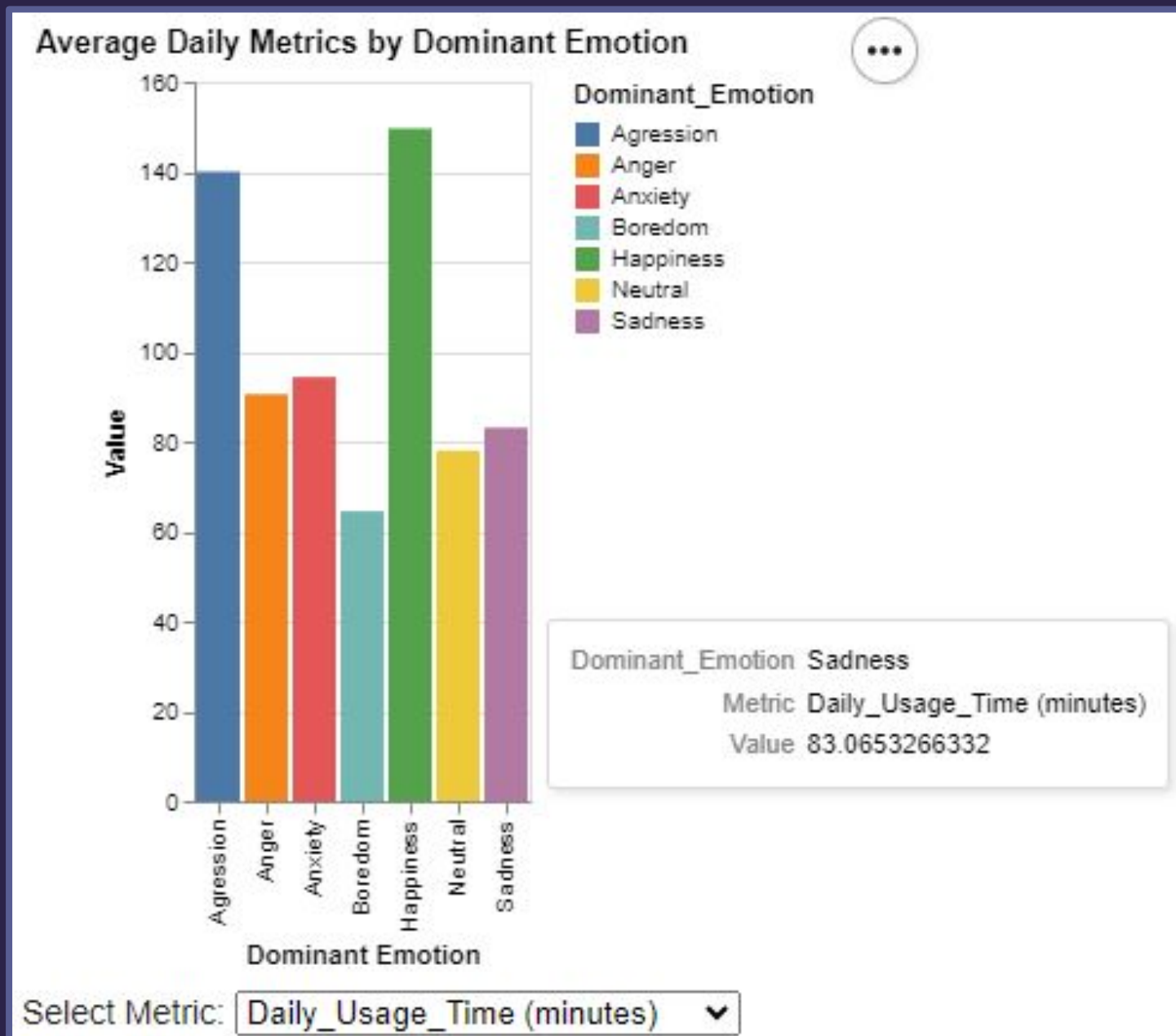
after



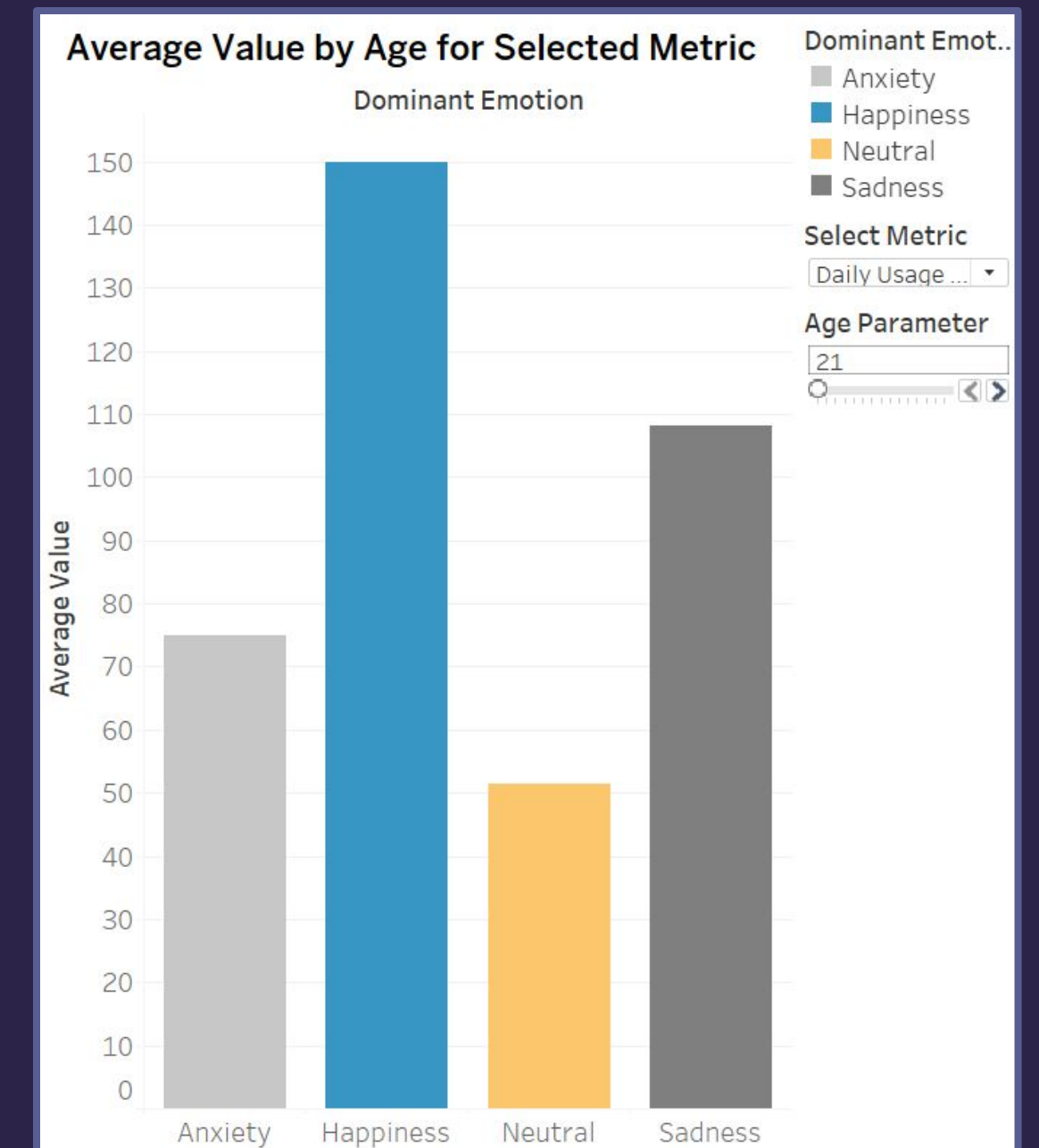
- all platforms are displayed for easier comparison
- which platform had highest usage is made clear

TASK 4 - INTERACTIONS

before

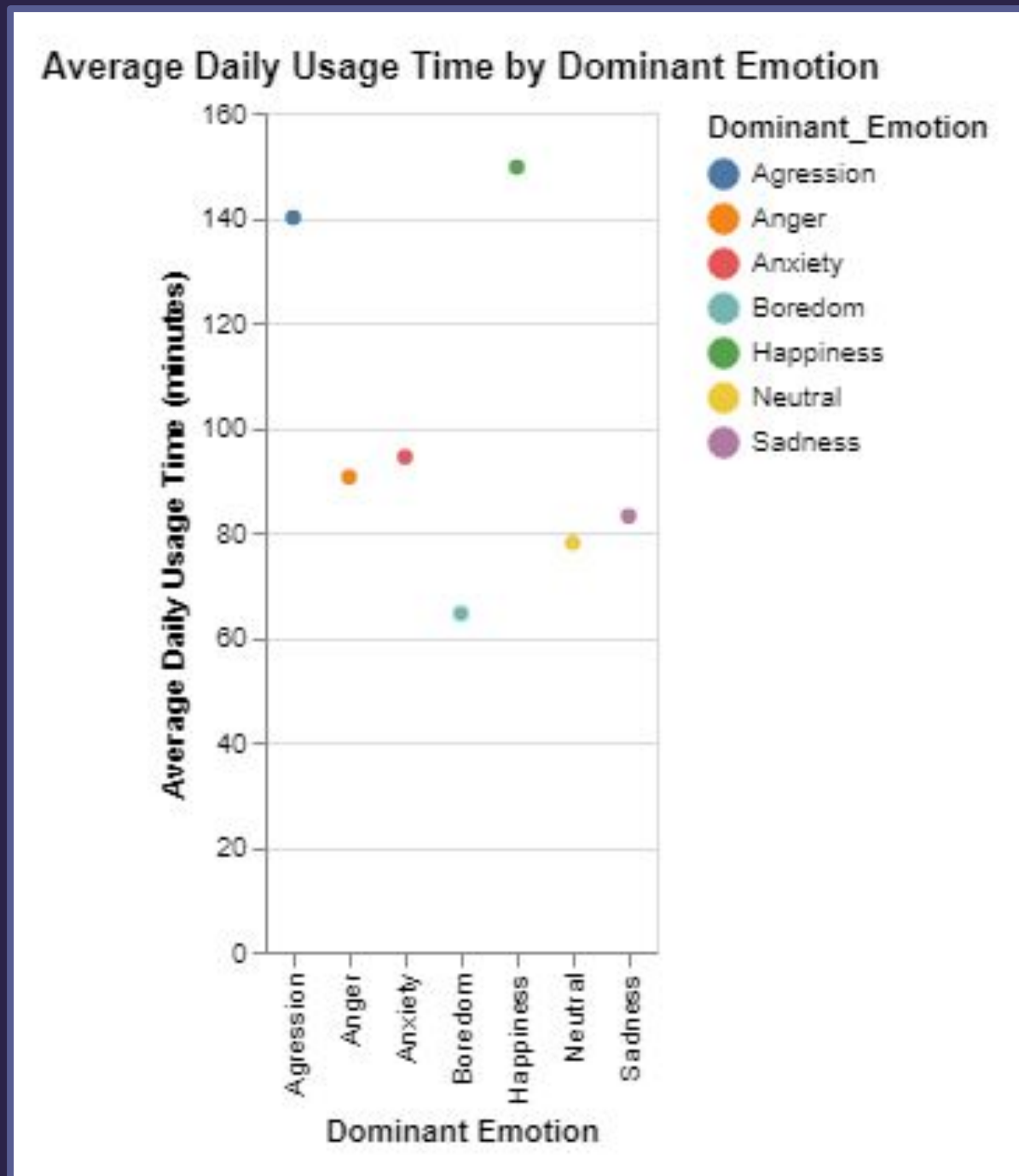


after

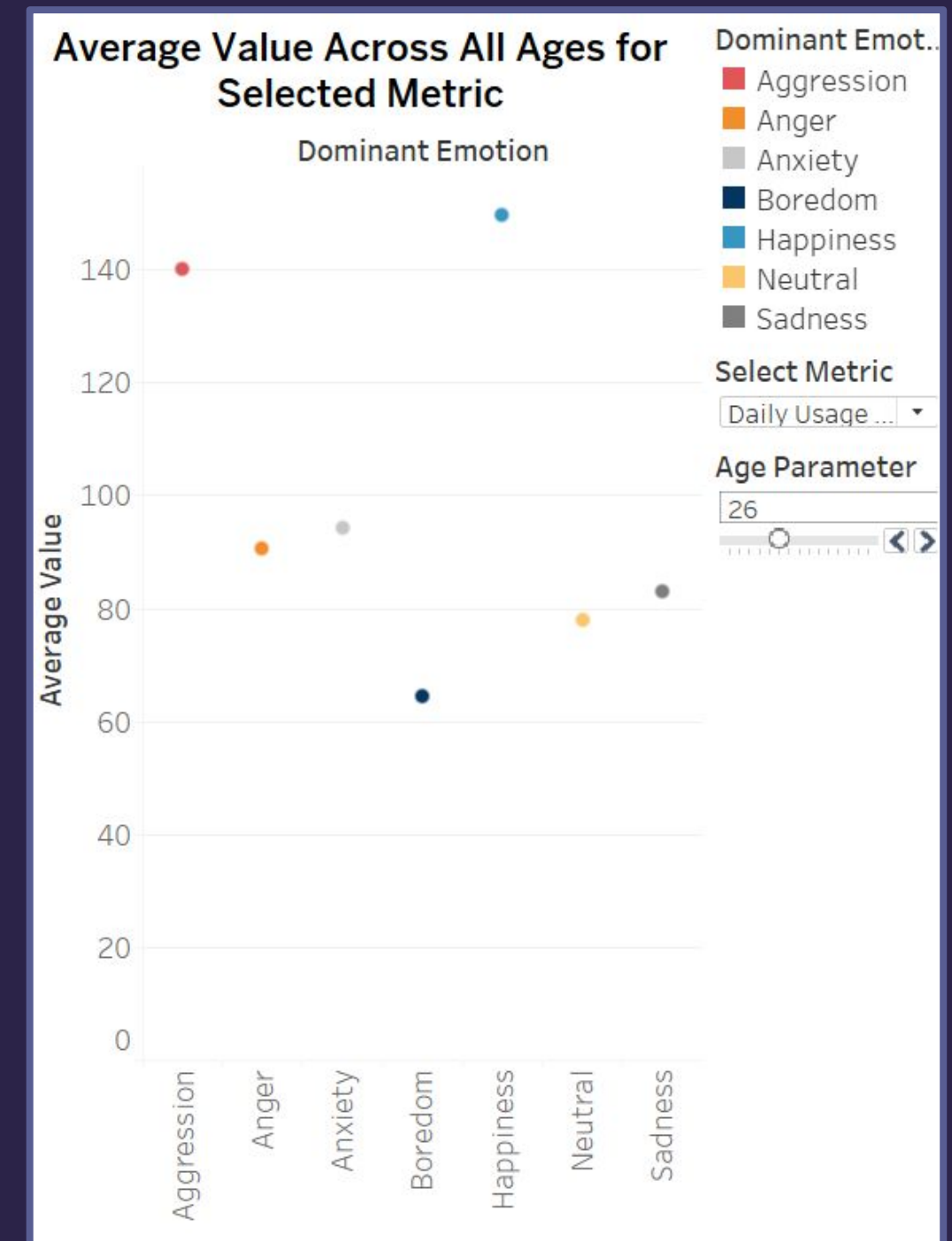


TASK 4 - INTERACTIONS

before

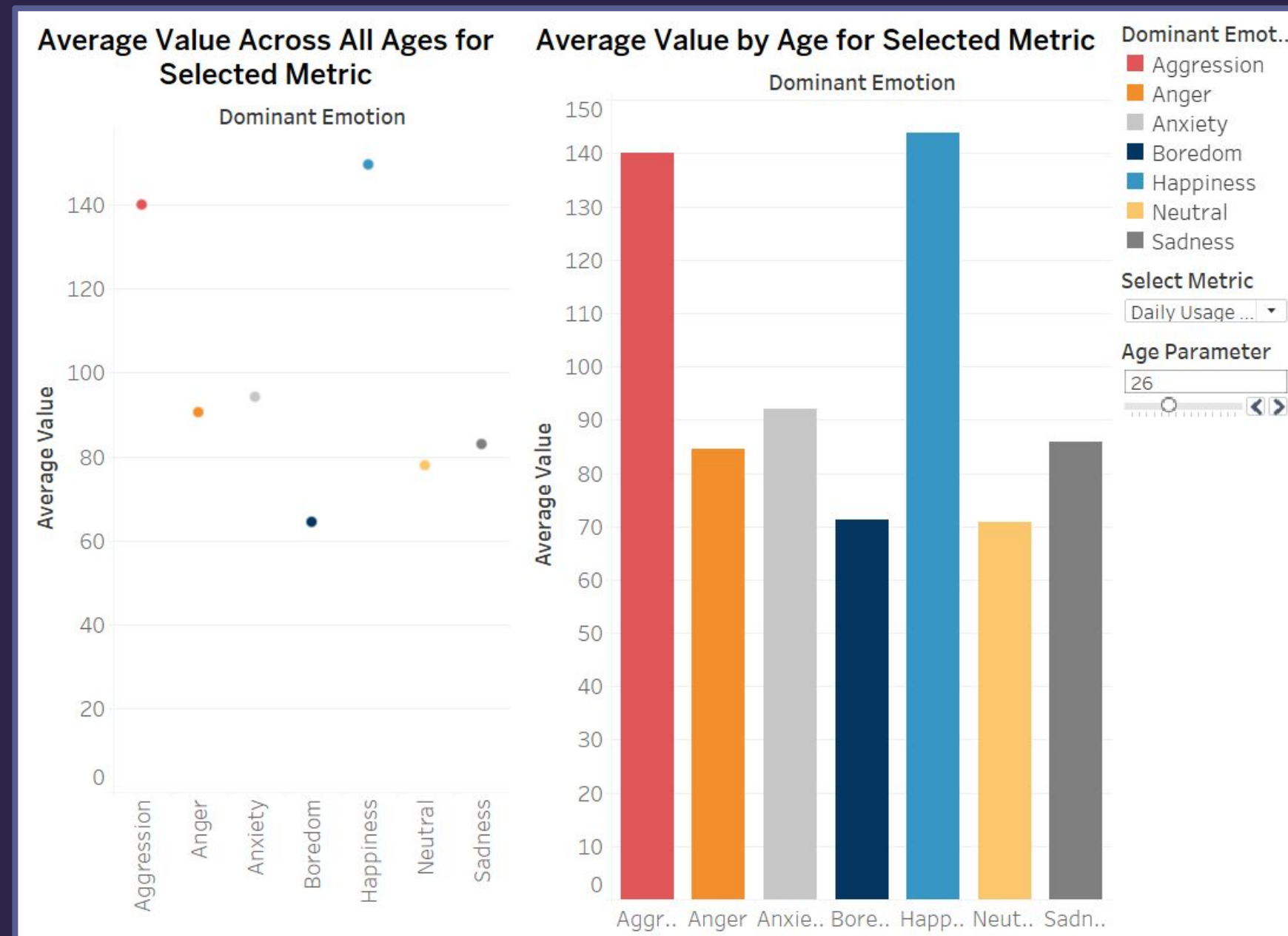


after



TASK 4 - INTERACTIONS

- **average value across all ages for selected metric**
 - shows dominant emotions of users across all ages by selected metric

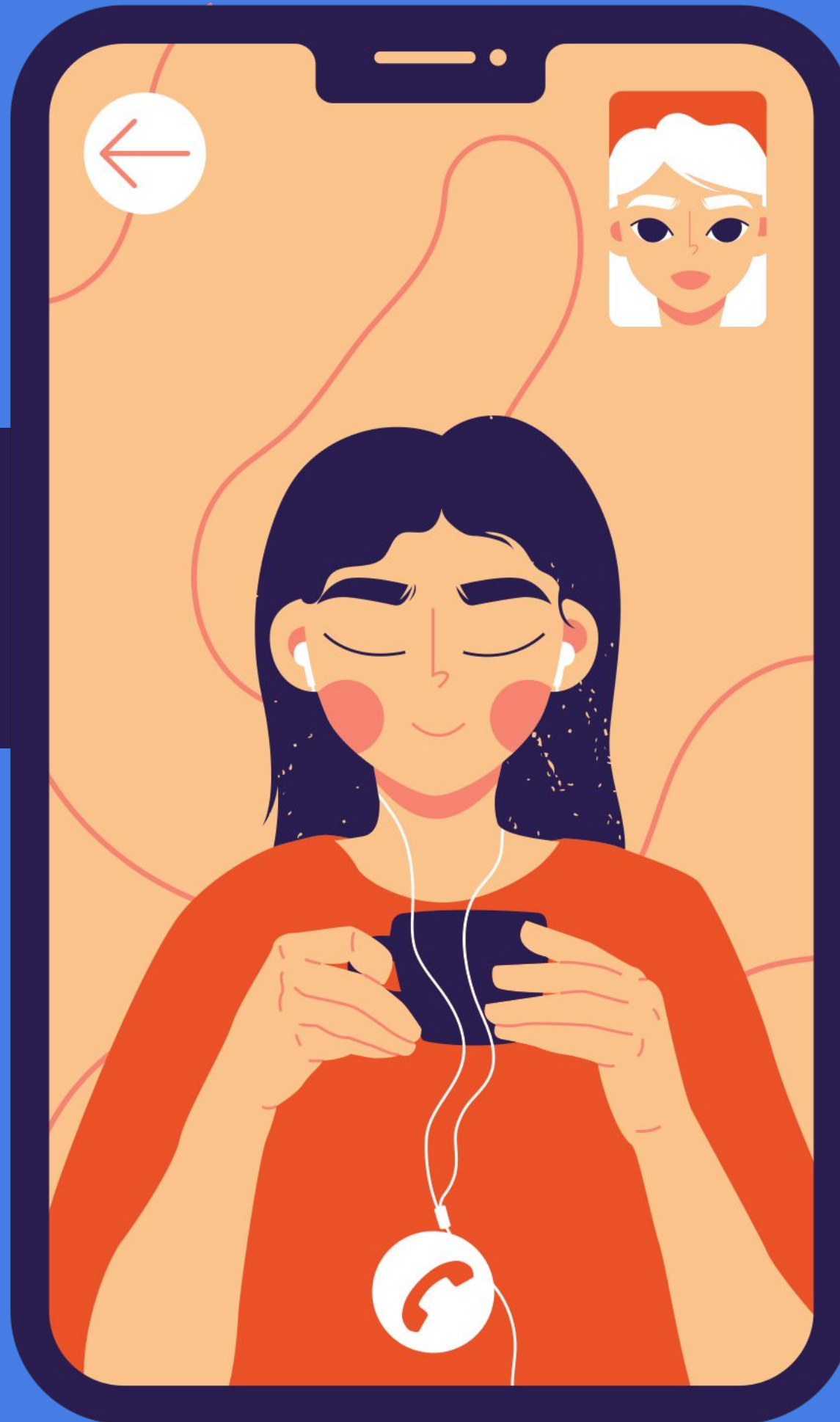


- **average value by age for selected metric**
 - displays dominant emotions for specific age group by selected metric

DEMO



THANK



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- myk5391@berkeley.edu

YOU

RESOURCE PAGE

Use these design resources in your Canva Presentation.

Fonts

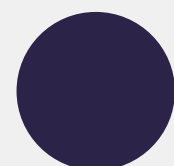
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uses the following free fonts:

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Design Elements

